

PDE4430 Mobile Robotics

Coursework

2025 – 2026 | Middlesex University Dubai

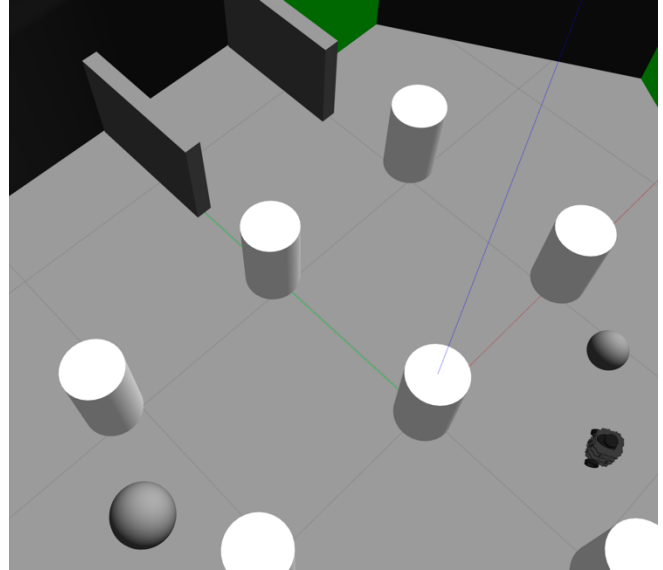
In this coursework assignment you will design a simple mobile robot to move spheres into the ‘pen’ formed by two walls.

The assessment uses the provided *assessment_world*. The robot needs to have a URDF file with all the necessary parameters defined to load into Gazebo in the simulation and be able to move and interact with objects in the simulation to complete the task or show that it could be completed. Instructions for the initial launching of the simulation are included in the readme file in the package.

You should use your git repository to store and submit your work. Create a ROS package to put your files in and regularly update it with clear commit messages to indicate what has been committed.

The submission needs the following:

1. The repository link containing the code and a README file containing information about the code, including the approach you have developed and instructions on running the project correctly.
2. 5-minute video presentation and demonstration of your work, including an overview of the solution and the project in action along with a voice-over explanation.



To achieve a **‘minimum’ pass** grade you will need to:

- Have a git repository that shows more than one commit over the life of the project.
- The git repository will contain the files needed to create your robot model and launch it in the gazebo simulation.
- On visual inspection the design should be appropriate to move the large sphere, and that it should be able to move the sphere to the target location.
- You demonstrate this in your video.

To achieve a **‘good’ pass**, in addition to the above, you will also need to produce:

- A simulated robot that will respond to appropriate ROS messages and move in the simulation.
- The spheres respond appropriately on coming in contact with the robot.
- Demonstrate the ability to move the large sphere in a way that clearly shows you could get it to the pen.
- Your design will have features that enable it to work well with more than one size of sphere.
- You may have some basic tools developed to aid teleoperation. This could include a sensor and simple ways to visualise it outside of Gazebo, and/or tools to control and monitor the robot.
- Your git repository shows regular commits as you worked on the project, with good separation of commits into independent logical parts.

To achieve an **‘excellent’ pass**, in addition to the previous things:

- You have started to develop strategies for autonomous operation - this may be that some information can be gathered that would enable this process, or you, or you have developed some area of functionality that would enable this to happen, but it may not be fully working.
- You may be able to identify where the balls are in code, but not move the robot autonomously, or move autonomously, but not identify the balls automatically, for example. To work on that you may be using more sophisticated features in git to manage multiple strands of independent work.

Additional notes:

- You are not to use the topics in Gazebo that provide information about location of items for core functionality – They may be used to test and verify things, but the running system should operate as if the robot was working in the real world, where most objects do not report their location to you.
- You can use other ROS nodes to get things done, but you need to be clear about what you are using, why you are using it, what functionality it gives you, and why it is a good fit in this situation.

This is an individual project, and there is no single 'correct' answer to what the robot is that would achieve this task. While robots are likely to have many similarities it is unlikely that your robots will be identical. The project will be marked on the following criteria:

25% Approach. Soundness and correctness of the chosen approach

25% Implementation. Appropriate and competent coding solutions; sensible program structure with functions and modularity; bugs and errors; safety checks; code clarity and readability.

25% Performance. Accuracy and reliability in the movement; robustness and generality; user interface and options; The code will be assessed based on how well the turtle (or turtles) move in the space, how much area they cover, and how quickly they manage to cover the entire area (Not based on velocity, but based on efficiency)

25% Documentation. Please include a detailed section on the design and architecture that your program follows. Describe the flow of information, including the nodes, topics, and messages that you have created. Feel free to also share the RQT graph of your project. In addition, the code should contain file and function descriptions and suitable line comments.

Due date: December 19, 2025